# Lab Report: Training dynamics of deep models

## 1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 50 epochs with a batch size of 128. The learning rate was set to 0.0003 with the Adam optimizer.

## 2 Results

Figure 1 shows the test accuracy during training.

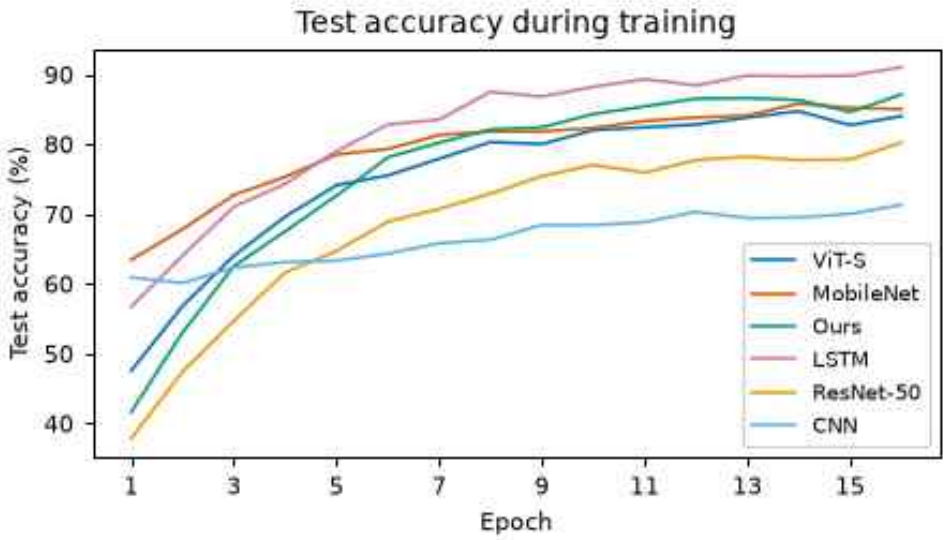



Figure 1: Test accuracy during training for the models in this experiment.

From epoch 9 to epoch 11, the test accuracy of MobileNet improves by a relative about 2%. LSTM reaches 60.5% test accuracy at epoch 15.

Figure 2 shows the test accuracy by dataset.

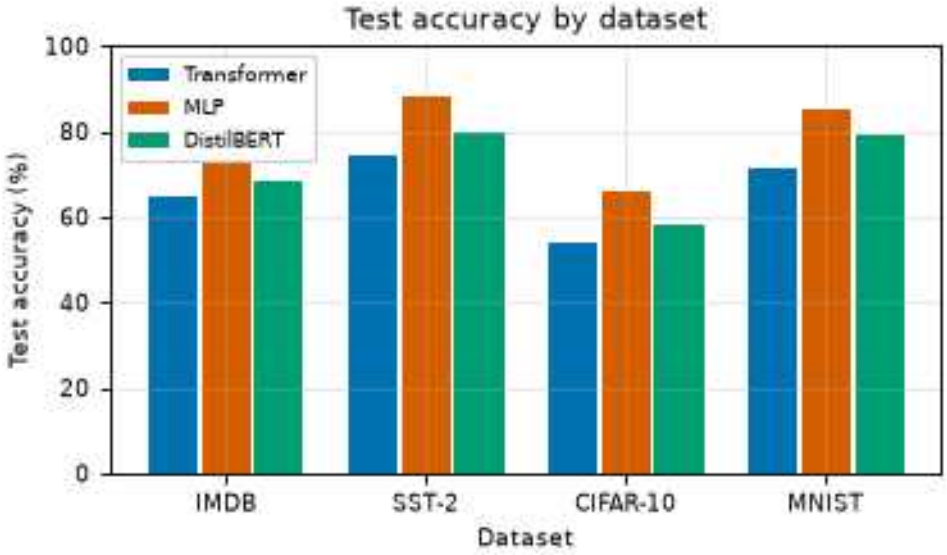



Figure 2: Test accuracy by dataset for the models in this experiment.

On CIFAR-10, MLP obtains a test accuracy of 66%. MLP achieves 79.0% test accuracy on MNIST.

Figure 3 shows the accuracy by dataset.

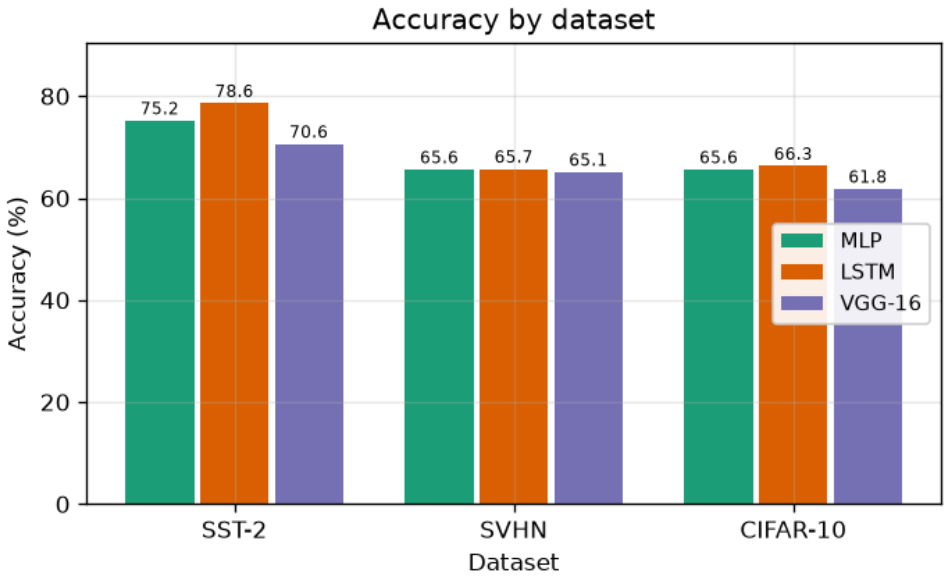



Figure 3: Accuracy by dataset for the models in this experiment.

On SST-2, LSTM is 8.0 points better than VGG-16. On CIFAR-10, VGG-16 obtains a accuracy of 63.6%.

Figure 4 shows the accuracy during training.

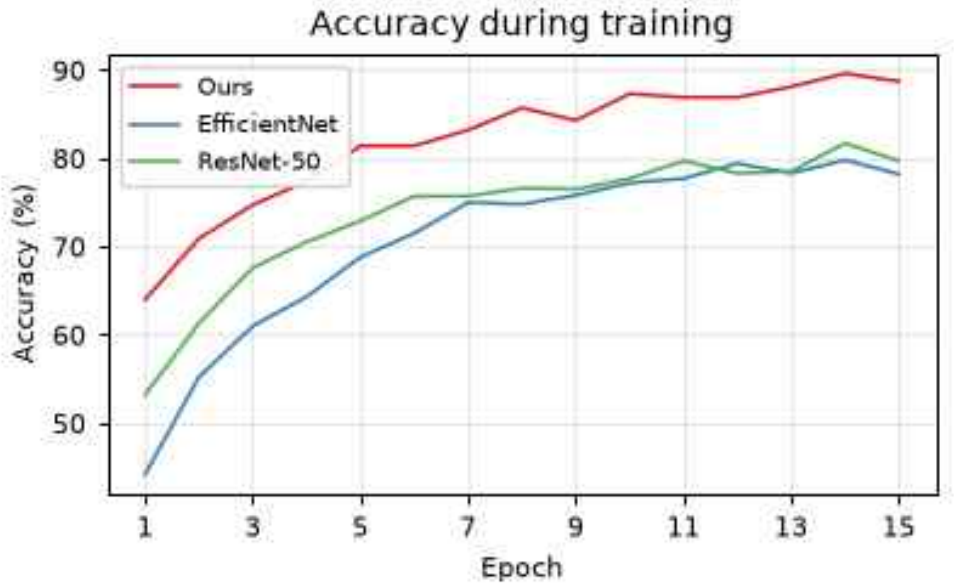



Figure 4: Accuracy during training for the models in this experiment.

EfficientNet reaches 46.7% accuracy at epoch 1. At epoch 2, Ours is 5.7 points better than EfficientNet. Ours reaches 89.8% accuracy at epoch 10 (Figure 4).

## 3 Discussion

On SVHN, SVM obtains an accuracy of 66.0%. Overall, the results are in line with what we expected from the literature.